

Efficient Verification of LTL via Transition Certificates

Verifying Linear Temporal Logic (LTL) specifications of discrete-time dynamical systems with continuous state spaces is challenging. Existing abstraction-free methods suffer from either conservatism or high computational cost. We propose transition certificates for LTL verification. Our approach transforms LTL verification into proving that accepting transitions occur finitely often. The transition persistence certificate employs ranking functions to prove accepting transitions occur finitely often. The transition safety certificate employs barrier certificates to prove accepting transitions are unreachable. Transition certificates reduce the search space and conservatism. We present an automated synthesis method based on neural network templates and demonstrate the effectiveness over some case studies.

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1 Introduction

Discrete-time dynamical systems are important mathematical models used to describe the characteristics of state changes in control systems. While these models can effectively capture system behavior, their infinite state space presents challenges for verification. Current methods for verifying temporal properties of these systems include the state-triplet method [23] and closure certificates [15]. However, these methods introduce new concerns: (i) The state-triplet method suffers from conservatism; (ii) The closure certificates have low synthesis efficiency. Developing a more efficient approach that reduces conservatism for verifying temporal properties in these systems remains a significant challenge.

Prior research has focused on formal verification of temporal properties—including safety, reachability, and reach-avoid specifications—within dynamical systems [2, 16, 24, 26, 27]. However, specifications for complex systems often require Linear Temporal Logic (LTL) [17] to capture temporal behaviors. Verifying LTL properties in control systems introduces several challenges. First, these properties describe behaviors across infinite execution trajectories [3], typically modeled via ω -automata. When employing proof certificates for verification, the state-triplet method requires unrolling all simple paths and blocking each individually. It is non-trivial to determine the number of certificates needed for synthesis and this method is conservative. Alternatively, closure certificates face the issue of enormous search spaces, as certificates are defined on $X \times Q \times X \times Q$. Finally, the infinite state space and nonlinear dynamics of control systems pose major obstacles for traditional verification methods.

In this work, we consider the formal verification of Linear Temporal Logic (LTL) properties for discrete-time dynamical systems using proof certificates. As an abstraction-free method, proof certificates have gained popularity in verification and static analysis domains [5–8]. Murali et al. [15] proposed closure certificates, proof certificates defined as real-valued functions over state pairs of dynamical systems, for persistence verification, which can be used to validate systems against LTL properties. Wongpiromsarn et al. [23] introduced the state-triplet method, which demonstrates that a system satisfies LTL properties by blocking all reachable paths. Inspired by these two methods, we propose transition safety certificates and transition persistence certificates. Their

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combined use enables verification of LTL specifications expressed via transition-based Nondeterministic Büchi Automata (NBA)[4]. Based on these certificates, we present an automated synthesis method using Satisfiability Modulo Theories (SMT) solvers. We show that combining these certificates reduces the conservatism of state-triplet methods while requiring smaller search spaces than closure certificates. We validate the effectiveness of our synthesis approach through two case studies.

Contributions. The main contributions of this paper are:

- We propose *transition certificates* for verifying LTL properties of discrete-time systems with continuous state spaces.
- We present a synthesis method to search for transition certificates within given neural network templates.
- We demonstrate how to use transition certificates to verify dynamical systems against LTL specifications with our case studies.

The organization of this paper is as follows: Section 2 covers preliminary knowledge. Section 3 introduces our proof certificates for LTL verification and compares with other methods. Section 4 presents the synthesis method. Section 5 demonstrates the effectiveness through case studies. Section 6 concludes the paper. The proofs of the relevant lemmas are provided in the appendix.

Related works: Verification of temporal logic properties via proof certificates has seen significant advancements. To validate discrete-time systems against LTL specifications, Wongpiromsarn et al. [23] developed a state-triplet method that uses barrier certificates to block all simple paths from initial states to accepting states in the automaton, requiring enumeration of state triplets (q_m, q'_m, q''_m) along these paths. Murali et al. [15] introduced closure certificates for persistence verification, which establish disjunctively well-founded transition invariants [18] for dynamical systems. Additionally, substantial research has addressed the verification of specific temporal behaviors such as reachability, safety, and avoidance[5–8]. Converting LTL specifications into Transition-based Generalized Büchi Automata (TGBA) typically yields smaller automata compared to Labeled Generalized Büchi Automata (LGBA) [9]. The degeneralization process [12] can convert a TGBA with multiple acceptance conditions into a Transition-based Büchi Automaton (TBA) featuring a single acceptance condition. Various tools exist for generating different automaton types from LTL formulas, such as Owl [14] and SPOT [10].

2 Preliminaries

2.1 Notation

We use $\mathbb{N}$ and $\mathbb{R}$ to represent the set of natural numbers and the set of real numbers, respectively. We define:

$$\begin{aligned}\mathbb{N}_{\sim i} &= \{n \in \mathbb{N} \mid n \sim i\}, \quad \text{where } i \in \mathbb{N}, \sim \in \{\geq, \leq, >, <\}, \\ \mathbb{R}_{\sim i} &= \{r \in \mathbb{R} \mid r \sim i\}, \quad \text{where } i \in \mathbb{R}, \sim \in \{\geq, \leq, >, <\}.\end{aligned}$$

Given a set S and a set S_1 , we define an indicator function:

$$I_S(x) = \begin{cases} 1, & \text{if } x \in S \\ 0, & \text{if } x \notin S \end{cases}$$

$|S|$ denotes the cardinality of S and $S \setminus S_1$ denotes $\{x \in S \mid x \notin S_1\}$.

Given a sequence $s = \langle s_0, s_1, \dots \rangle$, where $s_i \in S$ for $i \in \mathbb{N}$:

- $s[i]$ denotes s_i ;
- $s[i : j]$ denotes $\langle s_i, \dots, s_j \rangle$ for $j \geq i$;
- $s[i : \infty]$ denotes $\langle s_i, s_{i+1}, \dots \rangle$.

We say s visits S if there exists $i \in \mathbb{N}$ such that $s_i \in S$. $\text{Inf}(s)$ denotes the set of elements that appear in s infinitely often. S^ω denotes the set of infinite sequences over S , i.e., $S^\omega = \{s \mid s = \langle s_0, s_1, \dots \rangle \text{ and } s_i \in S \text{ for } i \in \mathbb{N}\}$. A function $f : S \rightarrow \mathbb{R}$ is bounded if and only if there exist $a, b \in \mathbb{R}$ such that $a \leq f(x) \leq b$ for all $x \in S$.

2.2 Discrete-time Dynamical System

A discrete-time dynamical system $\mathfrak{S}$ is a tuple (X, X_0, f) , where $X \subseteq \mathbb{R}^n$ represents the state set, $X_0 \subseteq X$ represents a set of initial states, and $f : X \rightarrow X$ represents the transition function. The evolution of the system state is described as follows:

$$x_{t+1} = f(x_t)$$

An infinite sequence of states $\langle x_0, x_1, x_2, \dots \rangle$ is called a *trajectory* of the system if $x_0 \in X_0$ and $x_{i+1} = f(x_i)$ for $i \in \mathbb{N}$. We use $\mathfrak{T}(\mathfrak{S})$ to represent all *trajectories* on $\mathfrak{S}$. The *predecessor* of x_{i+1} is x_i for $i \in \mathbb{N}$. In a given *trajectory*, x_{i+1} has only one *predecessor*. We use the notation x_i^{pre} to denote the *predecessor* of x_i in a *trajectory*.

2.3 Specification

This paper focuses on safety, persistence, and LTL properties of discrete-time dynamical systems.

Safety: A system $\mathfrak{S} = (X, X_0, f)$ is *safe* with respect to unsafe states $X_u \subseteq X$ if for any *trajectory* $\tau = \langle x_0, x_1, \dots \rangle \in \mathfrak{T}(\mathfrak{S})$, we have $x_i \notin X_u$ for all $i \in \mathbb{N}$.

Persistence: A system $\mathfrak{S} = (X, X_0, f)$ is *persistent* with respect to states $X_{VF} \subseteq X$ if for any *trajectory* $\tau = \langle x_0, x_1, \dots \rangle \in \mathfrak{T}(\mathfrak{S})$, elements of X_{VF} appear finitely often in τ .

Linear Temporal Logic (LTL): LTL formulae [17] are defined over a set of atomic propositions AP . A labelling function $\mathfrak{L} : X \rightarrow 2^{AP}$ maps system states to subsets of AP . Given a *trajectory* $\tau = \langle x_0, x_1, \dots \rangle \in \mathfrak{T}(\mathfrak{S})$, we denote $\mathfrak{L}(\tau) = \langle \mathfrak{L}(x_0), \mathfrak{L}(x_1), \dots \rangle$ as the corresponding *word*.

The syntax of LTL is:

$$\phi := \top \mid a \mid \neg\phi \mid \phi \wedge \phi \mid X\phi \mid \phi U \phi$$

where $a \in AP$ is an atomic proposition. Temporal operators finally (F) and always (G) are derived from next (X) and until (U).

Given a *word* $w = \mathfrak{L}(\tau)$ and an LTL formula ϕ , the semantics is defined inductively:

- $w \models \top$ always holds,
- $w \models a$ iff $a \in w[0]$,
- $w \models \phi_1 \wedge \phi_2$ iff $w \models \phi_1$ and $w \models \phi_2$,
- $w \models \neg\phi_1$ iff $w \not\models \phi_1$,
- $w \models X\phi_1$ iff $w[1 : \infty] \models \phi_1$,
- $w \models \phi_1 U \phi_2$ iff there exists $j \in \mathbb{N}$ such that $w[j : \infty] \models \phi_2$ and for all $i \in \mathbb{N}_{<j}$, $w[i : \infty] \models \phi_1$.

We say $\mathfrak{S} \models_{\mathfrak{L}} \phi$ if for any $\tau \in \mathfrak{T}(\mathfrak{S})$, we have $\mathfrak{L}(\tau) \models \phi$.

Nondeterministic Büchi Automaton (NBA): An NBA is a tuple $(\Sigma, Q, q_0, \delta, \text{Acc})$, where $\Sigma = 2^{AP}$ is the alphabet, Q is a finite set of states, $q_0 \in Q$ is the initial state, $\delta \subseteq Q \times \Sigma \times Q$ is the transition relation, and $\text{Acc} \subseteq \delta$ is the set of *accepting transitions*. Given a word $w = \langle w_0, w_1, \dots \rangle \in \Sigma^\omega$, a **run** on w is a sequence $r = \langle (r_0, w_0, r_1), (r_1, w_1, r_2), \dots \rangle$ where $r_0 = q_0$ and $(r_i, w_i, r_{i+1}) \in \delta$ for all $i \in \mathbb{N}$. An NBA accepts w if there exists a run where accepting transitions occur infinitely often, i.e., $\text{Inf}(r) \cap \text{Acc} \neq \emptyset$.

For an NBA $\mathcal{A} = (\Sigma, Q, q_0, \delta, \text{Acc})$, we define:

$$\text{Acc}^{\text{pair}} = \{(p, q) \mid \exists \sigma \in \Sigma. (p, \sigma, q) \in \text{Acc}\}, \quad (1)$$

$$\text{Acc}^{\text{start}} = \{p \mid \exists \sigma \in \Sigma, q \in Q. (p, \sigma, q) \in \text{Acc}\}, \quad (2)$$

$$\text{Acc}^{k,l} = \{(q_k, \sigma, q_l) \mid (q_k, \sigma, q_l) \in \text{Acc}\}. \quad (3)$$

We denote $\mathfrak{L}(\mathfrak{S}) = \{\mathfrak{L}(\tau) \mid \tau \in \mathfrak{T}(\mathfrak{S})\}$ as the *language* of $\mathfrak{S}$, and $\mathfrak{L}(\mathcal{A})$ as the language accepted by $\mathcal{A}$. To verify $\mathfrak{S} \models_{\mathfrak{L}} \phi$, we construct an NBA $\mathcal{A}$ for $\neg\phi$ and show $\mathfrak{L}(\mathfrak{S}) \cap \mathfrak{L}(\mathcal{A}) = \emptyset$.

2.4 Certificates

Barrier certificates and closure certificates are used to verify safety [19], persistence [15], and LTL properties [15], respectively. We provide their definitions and related conclusions from prior work.

Definition 2.1 (barrier certificate). Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ and $\mathcal{X}_u \subseteq \mathcal{X}$, a bounded function $B(x) : \mathcal{X} \rightarrow \mathbb{R}$ is a *barrier certificate* with respect to $\mathcal{X}_u$ such that $x_0 \in \mathcal{X}_0$, $x_u \in \mathcal{X}_u$, $x \in \mathcal{X}$ and $x' = f(x)$. We have:

$$B(x_0) \geq 0, \quad (4)$$

$$B(x_u) < 0, \quad (5)$$

$$B(x) \geq 0 \Rightarrow B(x') \geq 0. \quad (6)$$

THEOREM 2.2 (SAFETY). *The existence of a barrier certificate $B(x)$ implies that the system $\mathfrak{S}$ is safe with respect to $\mathcal{X}_u$.*

Definition 2.3 (closure certificate). Consider a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a set of atomic propositions AP , a labelling function $\mathfrak{L} : \mathcal{X} \rightarrow 2^{AP}$ and an NBA $\mathcal{A} = (2^{AP}, Q, q_0, \delta, Acc)$ represent $\neg\phi$. A bounded function $T : \mathcal{X} \times Q \times \mathcal{X} \times Q \rightarrow \mathbb{R}$ is a closure certificate for ϕ if there exists a value $\epsilon \in \mathbb{R}_{>0}$ such that for all states $x, y, y' \in \mathcal{X}$, $x' = f(x)$, states $q_i, q_j \in Q$, and $q'_i \in \delta(q_i, \mathfrak{L}(x))$, we have:

$$T((x, q_i), (x', q'_i)) \geq 0, \quad (7)$$

$$T((x', q'_i), (y, q_j)) \geq 0 \Rightarrow T((x, q_i), (y, q_j)) \geq 0. \quad (8)$$

and for all states $x_0 \in \mathcal{X}_0$, and $\ell, \ell' \in Acc$, we have:

$$\begin{aligned} T((x_0, q_0), (y, \ell)) \geq 0 \wedge T((y, \ell), (y', \ell')) \geq 0 \\ \Rightarrow T((x_0, q_0), (y', \ell')) \leq T((x_0, q_0), (y, \ell)) - \epsilon. \end{aligned} \quad (9)$$

THEOREM 2.4 (LTL). *Consider a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a set of atomic propositions AP , a labelling function $\mathfrak{L} : \mathcal{X} \rightarrow 2^{AP}$ and an LTL formula ϕ . Let an NBA $\mathcal{A} = (2^{AP}, Q, q_0, \delta, Acc)$ represent $\neg\phi$. The existence of a closure certificate satisfying conditions (7)-(9) implies that $\mathfrak{S} \models_{\mathfrak{L}} \phi$.*

2.5 Neural Network

A typical neural network consists of many interconnected neurons organized in a hierarchical structure. Each neuron serves as a unit that responds to weighted inputs received from other neurons and produces an output. Assume that $x \in \mathbb{R}^n$ represents the n -dimensional input vector of the neural network, $L - 1$ represents the number of hidden layers, and y is the scalar output of the neural network with respect to the input x . Thus, the feedforward neural network can be expressed with the following structure:

$$\begin{cases} z_l = \mathbf{W}_l x_{l-1} + \mathbf{b}_l, & 0 < l \leq L \\ x_l = \sigma(z_l), & 0 < l \leq L \\ y = x_L \end{cases} \quad (5)$$

where,

- z_l denotes the output vector of the linear operation in the l -th layer;
- $\mathbf{W}_l$ and $\mathbf{b}_l$ denote the weight matrix and bias vector for the l -th layer of the neural network, respectively;

- σ denotes the activation function. Common activation functions include the Tanh function, Sigmoid function, ReLU function.

2.6 Problem Statement

The search space for closure certificates is large, which makes the synthesis process take a long time. Next, we propose a new approach that improves on the search space.

Problem. Given a discrete-time dynamical system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a set of atomic propositions AP , a labeling function $\mathfrak{L} : \mathcal{X} \rightarrow 2^{AP}$, and an LTL formula ϕ , verify that $\mathfrak{S} \models_{\mathfrak{L}} \phi$ holds.

3 Transition Certificates

We propose transition certificates for LTL verification: *transition safety certificates* and *transition persistence certificates*. To verify that a system $\mathfrak{S}$ satisfies an LTL formula ϕ , we transform the verification problem into proving that accepting transitions in the NBA for $\neg\phi$ occur finitely often. By the definition of NBA acceptance, a word is accepted if and only if accepting transitions occur infinitely often along some run. If we prove all accepting transitions occur only finitely often for any trajectory in $\mathfrak{T}(\mathfrak{S})$, then no word in $\mathfrak{L}(\mathfrak{S})$ can be accepted by the NBA for $\neg\phi$, thereby proving $\mathfrak{S} \models_{\mathfrak{L}} \phi$. Both certificates ensure that accepting transitions in the NBA (representing $\neg\phi$) occur only finitely often. The transition safety certificate proves accepting transitions never occur by blocking reachability to their start states. The transition persistence certificate allows such states to be reachable but proves transitions occur finitely often via a ranking function.

3.1 Transition Safety Certificate

We prove accepting transitions never occur by showing Acc^{start} are unreachable in the product system.

Definition 3.1 (transition safety certificate). Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a labelling function $\mathfrak{L} : \mathcal{X} \rightarrow 2^{AP}$ and an NBA $\mathcal{A} = (2^{AP}, Q, q_0, \delta, Acc)$. Construct the product system $\mathfrak{S} \times \mathcal{A} = (\mathcal{Y}, \mathcal{Y}_0, F)$, where $\mathcal{Y} = \{(x, q) \mid x \in \mathcal{X}, q \in Q\}$, $\mathcal{Y}_0 = \{(x_0, q_0) \mid x_0 \in \mathcal{X}_0\}$, and $F(x, q) = \{(x', q') \mid x' = f(x) \wedge (q, \mathfrak{L}(x), q') \in \delta\}$. For $q_k \in Acc^{start}$, define the unsafe set $\mathcal{Y}_u = \{(x, q_k) \mid x \in \mathcal{X}\}$.

A bounded function $B_k(x, q) : \mathcal{X} \times Q \rightarrow \mathbb{R}$ is a *transition safety certificate* with respect to $q_k \in Acc^{start}$ if for all $(x, q) \in \mathcal{Y}$, $(x_0, q_0) \in \mathcal{Y}_0$, $(x', q') \in F(x, q)$, we have:

$$B_k(x_0, q_0) \geq 0, \quad (10)$$

$$B_k(x, q_k) < 0, \quad (11)$$

$$B_k(x, q) \geq 0 \Rightarrow B_k(x', q') \geq 0. \quad (12)$$

The certificate B_k guarantees q_k is unreachable, preventing all accepting transitions from q_k .

LEMMA 3.2. *If a transition safety certificate B_k with respect to $q_k \in Acc^{start}$ can be found, then all runs of words in $\mathfrak{L}(\mathfrak{S})$ never visit state q_k , thereby preventing all accepting transitions starting from q_k .*

3.2 Transition Persistence Certificate

We prove $Acc^{k,l}$ occur finitely often using a ranking function that decreases with each occurrence.

Definition 3.3 (transition persistence certificate). Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a labelling function $\mathfrak{L} : \mathcal{X} \rightarrow 2^{AP}$ and an NBA $\mathcal{A} = (2^{AP}, Q, q_0, \delta, Acc)$. For $(k, l) \in Acc^{pair}$, define the indicator function $I_{k,l}(x) = 1$ if $(q_k, \mathfrak{L}(x), q_l) \in Acc^{k,l}$, and 0 otherwise.

A bounded function $B_{k,l}(x) : \mathcal{X} \rightarrow \mathbb{R}$ is a *transition persistence certificate* with respect to $Acc^{k,l}$ if there exists $\epsilon > 0$ such that for all $x_0 \in \mathcal{X}_0$, $x \in \mathcal{X}$, $x' = f(x)$, we have:

$$B_{k,l}(x_0) \geq 0, \quad (13)$$

$$B_{k,l}(x) \geq 0 \Rightarrow B_{k,l}(x') \geq 0, \quad (14)$$

$$B_{k,l}(x) \geq B_{k,l}(x') + I_{k,l}(x)\epsilon. \quad (15)$$

The certificate $B_{k,l}$ is non-negative initially (13), remains non-negative along trajectories (14), and decreases by ϵ whenever the state's label matches an accepting transition from q_k to q_l (15). By (13) and (14), $B_{k,l}$ remains non-negative along any trajectory starting from $x_0 \in \mathcal{X}_0$. By (15), $B_{k,l}$ decreases by at least ϵ each time $I_{k,l}(x) = 1$. Since accepting transitions from q_k to q_l can only occur when $(q_k, \mathfrak{L}(x), q_l) \in Acc^{k,l}$, the number of such transitions is bounded by $B_{k,l}(x_0)/\epsilon$, ensuring finite occurrence.

LEMMA 3.4. *If a transition persistence certificate $B_{k,l}$ can be found, then all runs of words in $\mathfrak{L}(\mathfrak{S})$ make accepting transitions from q_k to q_l happen finitely often.*

3.3 LTL Verification

We now present the main theorem for LTL verification. For each accepting transition start state in the NBA, we use either a transition safety certificate or transition persistence certificates to prove accepting transitions from that state occur finitely often.

THEOREM 3.5 (LTL VERIFICATION). *Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a labelling function $\mathfrak{L} : \mathcal{X} \rightarrow \Sigma$ and an NBA $\mathcal{A} = (\Sigma, Q, q_0, \delta, Acc)$ representing the negation of an LTL formula ϕ . For each $q_k \in Acc^{start}$, if we can find either:*

- (1) *A transition safety certificate B_k for q_k , or*
- (2) *Transition persistence certificates $B_{k,l}$ for $Acc^{k,l}$ where $(k, l) \in Acc^{pair}$,*

then it can be guaranteed that $\mathfrak{S} \models_{\mathfrak{L}} \phi$.

PROOF. Given an accepting transition start state $q_k \in Acc^{start}$:

- (1) If a transition safety certificate B_k is found, by Lemma (3.2), all runs never visit state q_k , thereby preventing all accepting transitions starting from q_k .
- (2) If a transition persistence certificate $B_{k,l}$ is found, by Lemma (3.4), all runs make accepting transitions from q_k to q_l happen finitely often.

This ensures that all accepting transitions in Acc occur only finitely often. By the definition of NBA acceptance, a word w is accepted if and only if there exists a run where accepting transitions occur infinitely often, i.e., $Inf(r) \cap Acc \neq \emptyset$. Since all accepting transitions occur only finitely often, for any word $w \in \mathfrak{L}(\mathfrak{S})$ and any run r on w , we have $Inf(r) \cap Acc = \emptyset$. Therefore, no word in $\mathfrak{L}(\mathfrak{S})$ is accepted by automaton $\mathcal{A}$, implying $\mathfrak{L}(\mathfrak{S}) \cap \mathfrak{L}(\mathcal{A}) = \emptyset$. Since $\mathfrak{L}(\mathcal{A}) = \mathfrak{L}(\neg\phi)$, we have $\mathfrak{L}(\mathfrak{S}) \subseteq \mathfrak{L}(\phi)$, and consequently $\mathfrak{S} \models_{\mathfrak{L}} \phi$. $\square$

The two certificates are complementary: transition safety certificates prove that accepting transitions are unreachable, while transition persistence certificates prove that accepting transitions are reachable finitely. This combination reduces conservatism while maintaining smaller search spaces than closure certificates. To illustrate their complementarity, consider verifying $G\neg a$ with the NBA for Fa (Figure 1) and $Acc^{0,0} = \{(q_0, \emptyset, q_0), (q_0, \{a\}, q_0)\}$. For the transition persistence certificate $B_{0,0}$, conditions (13-15) require $B_{0,0}(x_0) \geq 0$ for $x_0 \in \mathcal{X}_0$, and $B_{0,0}(x) \geq B_{0,0}(x') + \epsilon$ for $(q_0, \mathfrak{L}(x), q_0) \in Acc^{0,0}$. It can be observed that, for any state $x \in \mathcal{X}$, the latter condition will always be satisfied, causing an infinite descent that violates the boundedness condition required by the transition persistence certificate; hence no valid certificate can be found. However, a transition safety certificate can prove q_0 is unreachable in the product system.

3.4 Compared with Other Methods

In the context of this discussion, we assume that the NBA for an LTL specification ϕ has already been constructed. The translation from an LTL formula to an NBA involves an exponential complexity with respect to the size of the formula [22]. The complexity of constructing the complement of an NBA is known to be EXPTIME [20].

The State-Triplet Approach. The state-triplet approach [23] uses barrier certificates (2.1) to separate safe regions $\mathcal{X}_{safe}$ from unsafe regions $\mathcal{X}_u$. It proves that accepting states are unreachable in the NBA for $\neg\phi$. Finding certificates for all such paths ensures $\mathfrak{L}(\mathfrak{S})$ never visits accepting states, verifying $\mathfrak{S} \models_{\mathfrak{L}} \phi$. Our transition safety certificate shares this reachability-blocking principle but operates on the product system $\mathfrak{S} \times \mathcal{A}$, eliminating the need to enumerate simple paths. The state-triplet method requires one certificate per triplet to block each path individually, while we synthesize one certificate per accepting transition start state. Transition persistence certificate further allows visiting accepting states while proving transitions occur finitely often. Combining both certificates reduces conservatism: beyond blocking reachability, we verify systems where runs reach accepting states yet satisfy the specification.

Closure Certificates. Closure certificates (2.3) are defined on $\mathcal{X} \times Q \times \mathcal{X} \times Q$, creating high-dimensional search spaces with heavy computational burdens. Our certificates operate on smaller spaces: transition persistence certificates on $\mathcal{X}$, transition safety certificates on $\mathcal{X} \times Q$. We require $|Acc^{pair}|$ transition persistence certificates and $|Acc^{start}|$ transition safety certificates, yielding a search space of at most $\mathcal{X} \times Q \times Q$ —smaller than $\mathcal{X} \times Q \times \mathcal{X} \times Q$. Case studies show that for benchmarks requiring closure certificates [15], our certificates synthesize in seconds with significant speedup.

4 Certificate Synthesis

This section presents an approach to synthesize transition certificates. Counterexample-Guided Inductive Synthesis (CEGIS) [21] has seen significant use in synthesizing barrier certificates. We thus provide a CEGIS approach to find transition certificates. Since neural networks have the capability to approximate arbitrary functions, using them as templates for certificates offers advantages over polynomials [25].

Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ and an NBA $\mathcal{A} = (2^{AP}, Q, q_0, \delta, Acc)$ represent $\neg\phi$. In the following, we describe the relevant conditions for the LTL objective ϕ . To find transition certificates as in Definitions 3.1 and 3.3, we first fix the certificate to be a neural network, as described in the Section 2.5.

The synthesis proceeds in two alternating phases: *candidate generation* and *verification*.

Candidate Generation. Based on certificate conditions, a loss function is designed for the training dataset. For transition safety certificates, the loss function is:

$$\begin{aligned} \mathcal{L}_B = & \sum_{(x,q) \in D_0} \text{ReLU}(-B_k(x, q; \theta)) \\ & + \sum_{(x,q) \in D_u} \text{ReLU}(B_k(x, q; \theta)) \\ & + \sum_{\substack{(x,q) \in D_{\mathcal{X}} \\ (q, \mathfrak{L}(x), q') \in \delta}} \text{ReLU}(B_k(x, q; \theta)) \cdot \text{ReLU}(-B_k(f(x), q'; \theta)), \end{aligned} \quad (16)$$

where $D_0, D_u, D_{\mathcal{X}}$ are the datasets sampled from $\mathcal{X}_0 \times \{q_0\}$, $\mathcal{X} \times \{q_k\}$, and $\mathcal{X} \times Q$, respectively. The terms in the loss function offer intuitive measures of the extent to which the neural network violates the certificate conditions.

For transition persistence certificates, the loss function is:

$$\mathcal{L}_B = \sum_{\substack{x \in D_X \\ (q_k, \mathfrak{Q}(x), q_l) \in Acc^{k,l}}} \text{ReLu}(B_{k,l}(f(x); \theta) + \epsilon - B_{k,l}(x; \theta)) \quad (17)$$

$$+ \sum_{\substack{x \in D_X \\ (q_k, \mathfrak{Q}(x), q_l) \notin Acc^{k,l}}} \text{ReLu}(B_{k,l}(f(x); \theta) - B_{k,l}(x; \theta)), \quad (18)$$

where $D_X \subseteq \mathcal{X}$ are the datasets sampled from $\mathcal{X}$.

To generate a neural certificate candidate, we employ gradient descent techniques to minimize the loss function. When the loss decreases to 0, it indicates that the learned neural network can be a candidate certificate. In addition, the activation functions differ: the transition safety certificate employs the ReLU activation function, while the transition persistence certificate uses the squared activation function. This design eliminates the need to verify non-negativity conditions during verification.

Verification. To verify the candidate neural certificate is valid, we formulate the negation of conditions and verify they are unsatisfiable.

For transition safety certificates, we check:

$$(x_0, q_0) \in \mathcal{Y}_0 \wedge \hat{B}_k(x_0, q_0; \hat{\theta}) < 0, \quad (19)$$

$$(x, q_k) \in \mathcal{Y}_u \wedge \hat{B}_k(x, q_k; \hat{\theta}) \geq 0, \quad (20)$$

$$(x, q) \in \mathcal{Y} \wedge (x', q') \in F(x, q) \wedge \hat{B}_k(x, q; \hat{\theta}) \geq 0 \wedge \hat{B}_k(x', q'; \hat{\theta}) < 0. \quad (21)$$

For transition persistence certificates, we verify:

$$x \in \mathcal{X} \wedge (q_k, \mathfrak{Q}(x), q_l) \notin Acc^{k,l} \wedge \hat{B}_{k,l}(x; \hat{\theta}) < \hat{B}_{k,l}(f(x); \hat{\theta}), \quad (22)$$

$$x \in \mathcal{X} \wedge (q_k, \mathfrak{Q}(x), q_l) \in Acc^{k,l} \wedge \hat{B}_{k,l}(x; \hat{\theta}) < \hat{B}_{k,l}(f(x); \hat{\theta}) + \hat{\epsilon}. \quad (23)$$

We choose dReal [11] as our SMT verifier, which ensures the correctness of its unsat decisions. Therefore, when dReal returns unsat for the given formula, it confirms that no solution exists within the specified precision, and the candidate certificate is valid. Otherwise, it means that dReal has found a counterexample that violates the certificate conditions. These counterexamples are added to the sample sets and the candidate generation phase is repeated. The process iterates until all verification queries are unsatisfiable (within the δ -completeness threshold of dReal), yielding a valid certificate.

This neural-network-based method does not depend on polynomial templates and can deal with general non-polynomial systems. However, its bottleneck primarily lies in the computation burden of SMT solving, especially when dealing with high-dimensional systems with highly nonlinear dynamics.

5 Experimental Evaluation

We evaluate our neural network-based certificate synthesis framework on two benchmark systems to demonstrate: (1) the effectiveness of transition safety and persistence certificates for LTL verification, (2) computational efficiency compared to prior closure certificate methods. All experiments run on Ubuntu 22.04.3 LTS (Intel i7-9750H, 8 GB RAM). We use dReal (precision $\delta = 0.0001$) for verification within the CEGIS framework. The candidate generation phase uses the Adam optimizer with learning rate 0.001 for 1000 iterations to minimize the loss function.

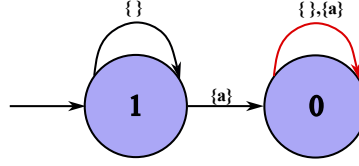


Fig. 1. NBA representing Fa (negation of safety specification). Accepting transitions (red) form a self-loop at q_0 .

5.1 Kuramoto Oscillator: Safety Verification

The Kuramoto model describes coupled oscillators in neuroscience and power systems [13]. We verify safety specifications $G\neg a$ for one-dimensional and two-dimensional scenarios.

5.1.1 One-Dimensional System. The system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$ has state space $\mathcal{X} = [0, 2\pi]$, initial states $\mathcal{X}_0 = [\frac{4\pi}{9}, \frac{5\pi}{9}]$, unsafe states $\mathcal{X}_u = [\frac{7\pi}{9}, \frac{8\pi}{9}]$, and dynamics:

$$f(x) = x + t_s \Omega + t_s K \sin(-x) - 0.532x^2 + 1.69, \quad (24)$$

where $t_s = 0.1$ (sampling time), $\Omega = 0.01$ (natural frequency), $K = 0.0006$ (coupling strength).

With $AP = \{a\}$ and labeling $\mathfrak{L}(x) = \{a\}$ iff $x \in \mathcal{X}_u$, safety is expressed as $G\neg a$. The negation Fa is represented by the NBA in Figure 1 with accepting transitions $(q_0, \{\}, q_0)$ and $(q_0, \{a\}, q_0)$ starting from $q_0 \in Acc^{start}$. We synthesize a transition safety certificate $B_0(x, q)$ using the CEGIS framework with a neural network template of architecture 2-10-1 (2 inputs for (x, q) , 10 hidden neurons with ReLU activation, 1 linear output). We collect 4 samples for D_0 , 7 for D_u , 14 for $D_{\mathcal{X}}$. CEGIS converges in 1 iteration.

5.1.2 Two-Dimensional System. The system extends to $\mathcal{X} = [0, \frac{8\pi}{9}]^2$, $\mathcal{X}_0 = [0, \frac{\pi}{9}]^2$, $\mathcal{X}_u = ([0, \frac{8\pi}{9}] \times [\frac{5\pi}{6}, \frac{8\pi}{9}]) \cup ([\frac{5\pi}{6}, \frac{8\pi}{9}] \times [0, \frac{8\pi}{9}])$, with dynamics:

$$f(x_1, x_2) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + t_s \begin{bmatrix} \Omega + 1.69 \\ \Omega + 1.69 \end{bmatrix} + K t_s \begin{bmatrix} \sin(x_2 - x_1) \\ \sin(x_1 - x_2) \end{bmatrix} - 0.532 t_s \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix}. \quad (25)$$

The labeling marks $\mathfrak{L}(x_1, x_2) = \{a\}$ when $x_1 \in [\frac{5\pi}{6}, \frac{8\pi}{9}]$ or $x_2 \in [\frac{5\pi}{6}, \frac{8\pi}{9}]$, using the same NBA from Figure 1. We synthesize a transition safety certificate $B_0(x_1, x_2, q)$ using the CEGIS with neural network template of architecture 3-10-1 (3 inputs for (x_1, x_2, q) , 10 hidden neurons with ReLU activation, 1 output). We collect 4 samples for D_0 , 9 for D_u , 18 for $D_{\mathcal{X}}$. The framework converges in 1 iteration.

5.2 Two-Room Temperature: Persistence Verification

We verify an interconnected temperature control system [1]. State space $\mathcal{X} = [20, 34]^2$ is the set of system states, $\mathcal{X}_0 = [21, 24]^2$ is the set of initial states, $\mathcal{X}_{VF} = [20, 26]^2$ is the set of finitely visited states, and:

$$f(x_1, x_2) = A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \mu T_h \begin{bmatrix} u(x_1) \\ u(x_2) \end{bmatrix} + \theta \begin{bmatrix} T_e \\ T_e \end{bmatrix}, \quad (26)$$

where

$$A = \begin{bmatrix} 1 - 2\alpha - \theta - \mu u(x_1) & \alpha \\ \alpha & 1 - 2\alpha - \theta - \mu u(x_2) \end{bmatrix}, \quad (27)$$

with parameters $\alpha = 0.004$, $\theta = 0.01$, $\mu = 0.15$, $u(x_i) = 0.59 - 0.011x_i$, $T_h = 40$, $T_e = 0$.

With $AP = \{a, b\}$, label $\mathfrak{L}(x_1, x_2) = \{a\}$ for $(x_1, x_2) \in \mathcal{X}_0$, $b \in \mathfrak{L}(x_1, x_2)$ for $(x_1, x_2) \in \mathcal{X}_{VF}$. The LTL specification is $a \Rightarrow FG\neg b$. The NBA represents $a \wedge GFb$ (the negation of the LTL specification) in Figure 2. We simultaneously synthesize the transition safety certificate $B_1(x_1, x_2, q)$ for q_1 and the transition persistence certificate $B_{1,1}(x_1, x_2)$

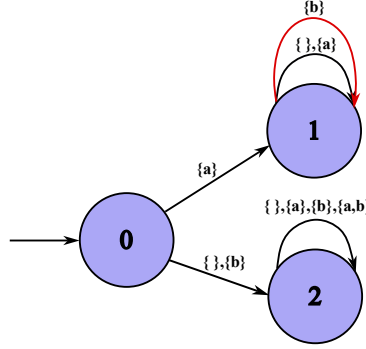


Fig. 2. NBA for $a \wedge GFb$ (negation of $a \Rightarrow FG\neg b$). Accepting transitions (red) are $(q_1, \{b\}, q_1)$ and $(q_1, \{a, b\}, q_1)$.

for $Acc^{1,1}$. We synthesize $B_0(x_1, x_2, q)$ using the CEGIS with neural network template of architecture 3-10-1 (3 inputs for (x_1, x_2, q) , 10 hidden neurons with ReLU activation, 1 output). We synthesize $B_{1,1}(x_1, x_2)$ using the CEGIS with neural network template of architecture 2-3-1 (2 inputs for (x_1, x_2) , 3 hidden neurons with squared activation, 1 output). For the transition persistence certificate, we collect 100 samples for D_X . For the transition safety certificate, we collect 4 samples for D_0 , 9 for D_u , 18 for D_X . We successfully obtain a transition persistence certificate.

5.3 Comparison

Table 1 compares synthesis times. Our neural network certificates achieve speedup over closure certificates, confirming the efficiency gains from reduced search space complexity.

Table 1. Synthesis time comparison (seconds).

Benchmark	Our Method	Closure [15]	Speedup
Kuramoto 1D	2.8	600	71×
Kuramoto 2D	3	6600	2200×
Two-Room	1632	5400	3.3×

6 Conclusion

This paper introduces transition certificates for verifying LTL properties of discrete-time dynamical systems. They reduce conservatism compared to state-triplet methods [23] and have search spaces no larger than $X \times Q \times Q$, significantly smaller than closure certificates [15]. We present a neural network-based synthesis method that achieves a speedup over closure certificates. Future work will focus on improving certificate synthesis efficiency and investigating its application to controller synthesis.

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A Proof

THEOREM A.1. *Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, a trajectory $\tau = \langle x_0, x_1, \dots \rangle \in \mathfrak{T}(\mathfrak{S})$, a labelling function $\mathfrak{L} : \mathcal{X} \rightarrow \Sigma$ and an NBA $\mathcal{A} = (\Sigma, Q, q_0, \delta, \text{Acc})$, if a transition safety certificate $B_k(x, q)$ with respect to $q_k \in \text{Acc}^{\text{start}}$ can be found, then it can be guaranteed that all runs of all words in $\mathfrak{L}(\mathfrak{S})$ cannot visit the state q_k .*

PROOF. By Definition 3.1, we have $\mathcal{Y}_u = \{(x, q_k) \mid x \in \mathcal{X}\}$ as the unsafe set in the product system $\mathfrak{S} \times \mathcal{A}$. We verify that B_k satisfies the barrier certificate conditions (4, 5, 6) with respect to $\mathcal{Y}_u$:

- Condition (10) states $B_k(x_0, q_0) \geq 0$ for all $(x_0, q_0) \in \mathcal{Y}_0$, which corresponds to condition (4).
- Condition (11) states $B_k(x, q_k) < 0$ for all $(x, q_k) \in \mathcal{Y}_u$, which corresponds to condition (5).
- Condition (12) states $B_k(x, q) \geq 0 \Rightarrow B_k(x', q') \geq 0$ for all $(x', q') \in F(x, q)$, which corresponds to condition (6).

By Corollary (2.2), any trajectory of the product system $\mathfrak{S} \times \mathcal{A}$ cannot visit $\mathcal{Y}_u$.

Now we establish the connection to automaton runs. For any word $w = \mathfrak{L}(\tau) \in \mathfrak{L}(\mathfrak{S})$ where $\tau = \langle x_0, x_1, \dots \rangle$ is a trajectory of $\mathfrak{S}$, the corresponding run r of w on $\mathcal{A}$ is a sequence $r = \langle (r_0, w_0, r_1), (r_1, w_1, r_2), \dots \rangle$ where $r_0 = q_0$. The sequence of product states $\langle (x_0, r_0), (x_1, r_1), \dots \rangle$ forms a trajectory in $\mathfrak{S} \times \mathcal{A}$. Since this trajectory cannot visit $\mathcal{Y}_u = \{(x, q_k) \mid x \in \mathcal{X}\}$, we have $r_i \neq q_k$ for all i . Therefore, runs of all words in $\mathfrak{L}(\mathfrak{S})$ cannot visit state q_k . $\square$

THEOREM A.2. *Given a system $\mathfrak{S} = (\mathcal{X}, \mathcal{X}_0, f)$, an NBA $\mathcal{A} = (\Sigma, Q, q_0, \delta, \text{Acc})$, and $(k, l) \in \text{Acc}^{\text{pair}}$, if a transition persistence certificate $B_{k,l}$ can be found, then it can be guaranteed that all runs of the words in $\mathfrak{L}(\mathfrak{S})$ make accepting transitions from q_k to q_l happen finitely often.*

PROOF. We define $\mathcal{X}_{k,l} = \{x \mid (q_k, \mathfrak{L}(x), q_l) \in \text{Acc}^{k,l}\}$ as the set of states where accepting transitions from q_k to q_l can occur.

Consider any trajectory $\tau = \langle x_0, x_1, \dots \rangle \in \mathfrak{T}(\mathfrak{S})$ and its corresponding word $w = \mathfrak{L}(\tau)$. Let r be a run of w on $\mathcal{A}$ starting from q_0 . An accepting transition from q_k to q_l can occur at step i only if $(q_k, \mathfrak{L}(x_i), q_l) \in \text{Acc}^{k,l}$, which is equivalent to $x_i \in \mathcal{X}_{k,l}$. Therefore, the number of accepting transitions from q_k to q_l is bounded by the number of times the trajectory visits $\mathcal{X}_{k,l}$.

By Definition 3.3, we have:

- $B_{k,l}(x_0) \geq 0$ by condition (13).
- $B_{k,l}(x_i) \geq 0$ for all i by condition (14), since $B_{k,l}$ remains non-negative along the trajectory.
- $B_{k,l}(x_i) \geq B_{k,l}(x_{i+1}) + I_{k,l}(x_i)\epsilon$ by condition (15), where $I_{k,l}(x_i) = 1$ if $x_i \in \mathcal{X}_{k,l}$ and 0 otherwise.

Suppose the trajectory visits $\mathcal{X}_{k,l}$ at indices $i_1, i_2, \dots, i_m$ where m is the number of times accepting transitions occur. At each visit, $B_{k,l}$ decreases by at least ϵ :

$$B_{k,l}(x_0) \geq B_{k,l}(x_{i_1}) + \epsilon \geq B_{k,l}(x_{i_2}) + 2\epsilon \geq \dots \geq B_{k,l}(x_{i_m}) + m\epsilon. \quad (28)$$

Since $B_{k,l}(x_{i_m}) \geq 0$ by condition (14), we have $B_{k,l}(x_0) \geq m\epsilon$, which gives $m \leq \frac{B_{k,l}(x_0)}{\epsilon}$. Therefore, the number of accepting transitions from q_k to q_l is bounded, ensuring they occur only finitely often. $\square$